

Inverses from Graphs and Tables - Practice

Algebra II | Unit 1 | Lesson 10 Practice | TEK A2.2.B

Objective: I can find an inverse from pairs or a table, and predict whether an inverse will be a function.**Section 1 — Recall**

1. Given $f = \{ (2, 9), (5, 1), (7, -3) \}$, write f^{-1} .

$$f^{-1} = \{ \underline{\hspace{2cm}} \}$$

2. If $(4, -6)$ is on g , name the matching point on g^{-1} .

Section 2 — Apply

3. Complete the inverse table for f : $x = -3, 1, 6 \rightarrow f(x) = 8, 2, -5$. Give f^{-1} as a table.

$$f^{-1}: x = 8, 2, -5 \rightarrow f^{-1}(x) = \underline{\hspace{2cm}}$$

4. The domain of $h = \{ (0, 3), (4, 3), (9, -2) \}$ is $\{0, 4, 9\}$. What is the range of h^{-1} ?

$$\text{Range of } h^{-1}: \{ \underline{\hspace{2cm}} \}$$

Section 3 — Challenge

5. A table gives k : $(1, 7), (3, 7), (5, 2)$. Write k^{-1} , then decide whether it is a function, explaining why.

$$k^{-1} = \{ \underline{\hspace{2cm}} \}$$

A function? _____

6. A graph passes through $(-2, 5)$, $(0, 5)$, and $(3, 8)$. Without seeing the full picture, can you tell whether its inverse will be a function? Explain.

Section 4 — Explain It

7. A classmate says: "To find an inverse, you just flip the graph upside down." Is this description correct? Explain what actually happens geometrically, and why "upside down" isn't quite right.
